



Spiking neural networks for physiological and speech signals: a review

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Abstract

The integration of Spiking Neural Networks (SNNs) into the analysis and interpretation of physiological and speech signals has emerged as a groundbreaking approach, offering enhanced performance and deeper insights into the underlying biological processes. This review aims to summarize key advances, methodologies, and applications of SNNs within these domains, highlighting their unique ability to mimic the temporal dynamics and efficiency of the human brain. We dive into the core principles of SNNs, their neurobiological underpinnings, and the computational advantages they bring to signal processing, particularly in handling the temporal and spatial complexities inherent in physiological and speech data. Comparative analyses with conventional neural network models are presented to underscore the superior efficiency, lower power consumption, and higher temporal resolution of SNNs. The review further explores challenges and future prospects, highlighting the potential of SNNs to revolutionize wearable healthcare monitoring systems, neuroprosthetic devices, and natural language processing technologies. By providing a comprehensive overview of current strategies, this review aims to inspire innovative approaches in the field, fostering advances in real-time and energy-efficient processing of complex biological signals.

Keywords Spiking neural networks · Neuron models · Spike encoding · Learning · Physiological signals · Speech

1 Introduction

The advent of Spiking Neural Networks (SNNs) represents a paradigm shift in the field of computational neuroscience and artificial intelligence, propelling us closer to the elusive goal of mirroring the computational efficiency and dynamism of the human brain. Unlike traditional artificial neural networks (ANNs) that process information in a continuous manner, SNNs incorporate discrete events or 'spikes', which are more akin to the neuronal activity observed in biological systems. This fundamental difference enables SNNs to process information in a way that is not only more biologically plausible but also more energy-efficient, especially when dealing with temporal data such as physiological and speech signals [1, 2].

The analysis of physiological signals-ranging from the electrical activity of the brain (EEG) to the heart (ECG) and muscle activity (EMG)-is crucial for a wide array of

medical diagnostics and therapeutic interventions. Traditional approaches often rely on signal processing techniques that, while powerful, can struggle to fully capture the dynamic and often non-linear nature of these physiological processes. SNNs, with their intrinsic ability to model temporal dynamics, offer a compelling alternative. They can not only provide more accurate analyses by capturing the rich temporal information contained within these signals but also lead to more effective real-time monitoring and intervention strategies, with applications extending into the realms of seizure detection, cardiac anomaly identification, and even in the nuanced control of neuroprosthetics [3, 4].

In the domain of speech processing, the potential of SNNs is equally transformative. Speech is inherently temporal and carries a wealth of information in its temporal variations. Conventional speech recognition systems, despite their sophistication, often fall short in accurately modeling these variations, particularly under noisy or varied conditions. SNNs, by leveraging their temporal precision and event-driven nature, hold the promise of significantly enhancing the accuracy and robustness of speech recognition systems. They are particularly adept at capturing the subtle nuances of speech, such as intonation, stress, and rhythm, thereby

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paving the way for more natural and efficient human-computer interactions [5]

The integration of SNNs into the analysis of physiological and speech signals, however, is not devoid of challenges. The complexity of designing and training SNNs, especially given their non-differentiable nature, has necessitated the development of novel learning algorithms and training paradigms. Moreover, the computational demands of SNNs, despite their energy efficiency in theory, require specialized neuromorphic hardware for practical, large-scale applications. This has led to significant advancements in neuromorphic engineering, with developments such as Intel's Loihi and IBM's TrueNorth chips, which are designed to efficiently simulate large-scale SNNs [6].

Furthermore, the field of SNNs in physiological and speech signal processing is rapidly evolving, with ongoing research exploring the integration of SNNs with other computational models, the development of hybrid systems, and the application of SNNs in novel contexts such as biofeedback systems and affective computing. These endeavors not only expand the utility of SNNs but also contribute to our understanding of the underlying biological processes, offering insights into human cognition and neural functioning [7].

This review article aims to delve deeper into the transformative impact of SNNs on the analysis of physiological and speech signals. By providing a comprehensive overview of the foundational principles of SNNs, their neurobiological inspirations, and their computational advantages, we set the stage for an in-depth exploration of their applications in these domains. Through a detailed examination of current research, challenges, and future directions, we aim to highlight the potential of SNNs to not only advance our technological capabilities but also to provide deeper insights into the complex workings of the human brain and body (Fig. 1).

2 Spiking neuron models

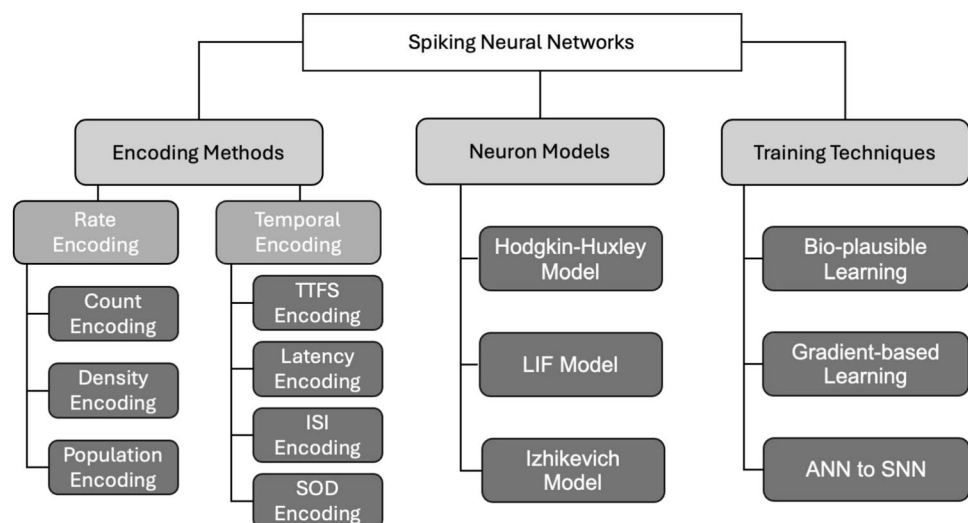
The quest to understand and replicate the computational capabilities of the human brain has led to the development of various spiking neuron models. These models aim to capture the essence of neuronal dynamics through mathematical formulations, enabling the simulation of neural networks that more closely resemble biological systems. In this section, we explore three pivotal spiking neuron models: the Hodgkin-Huxley (HH) model, the Leaky Integrate-and-Fire (LIF) model, and the Izhikevich model. Each of these models offers a different level of complexity and biological fidelity, making them suitable for various applications in computational neuroscience and neuromorphic engineering.

2.1 Hodgkin–Huxley model

The Hodgkin-Huxley model stands as a cornerstone in the field of neurophysiology, offering a detailed mathematical description of how action potentials are initiated and propagated in neurons [8]. Developed from experiments on the squid giant axon, this model captures the intricate dance of sodium and potassium ions across the neuronal membrane, driven by voltage-gated ion channels. The model's differential equations describe the dynamics of membrane potential changes in response to ionic currents, providing insights into the biophysical processes underlying neural excitability and signaling [9].

While the Hodgkin-Huxley model's comprehensive detail allows for an accurate representation of neuronal action potentials, its complexity renders it computationally intensive, particularly when simulating large networks of neurons. Despite this, the model's depth has facilitated a profound understanding of the electrophysiological

Fig. 1 Overview of this article's structure



properties of neurons, influencing the development of drugs targeting ion channels and contributing to the field of computational neuroscience by setting a benchmark for model accuracy [10, 11].

2.2 Leaky integrated and fire (LIF) model

The LIF model offers a more abstract representation of neuronal activity, focusing on the integration of synaptic inputs and the generation of action potentials upon reaching a specific threshold. This model simplifies the neuron to a circuit-like element, with the ‘leak’ term representing the passive decay of membrane potential towards a resting state, akin to a capacitor discharging through a resistor. This abstraction allows the LIF model to efficiently simulate spiking activity across large neural networks, making it a popular choice in various applications, from theoretical studies of neural coding to practical implementations in neuromorphic systems.

Despite its simplicity, the LIF model captures essential features of neuronal excitability and has been extended in various ways to include features like synaptic dynamics, refractory periods, and adaptive thresholds. These extensions enhance the model’s utility in studying neural computation and network dynamics, making it a versatile tool in the exploration of how neural systems process information [12–15].

2.3 Izhikevich model

The Izhikevich model represents a synthesis of biological realism and computational efficiency, capable of reproducing the diverse array of spiking and bursting patterns observed in real neurons [2]. By distilling the complex dynamics of neuronal activity into a two-dimensional system of ordinary differential equations, this model achieves a balance that allows for the simulation of large neural networks with rich dynamical behaviors. The model’s versatility is evident in its ability to replicate various types of neuronal firing patterns through the adjustment of a few parameters, making it an invaluable tool for both theoretical analyses and practical applications.

The Izhikevich model’s capacity to mimic different neuronal behaviors with computational efficiency has made it a preferred choice for large-scale simulations of brain dynamics and for implementation in neuromorphic hardware, where the fidelity of neural emulation is crucial. Its development has spurred further research into the mechanisms underlying neural excitability and plasticity, contributing to our understanding of neural computation and the pathophysiology of neurological disorders [16].

3 Spike encoding

SNNs employ “all-or-nothing” spike pulses to encode and deliver stimuli. These spike pulses are produced in response to an action potential within the human nervous system. Spike encoding is a fundamental aspect of how SNNs process and transmit information, distinguishing them from traditional ANNs. This section delves into the principles of spike encoding, focusing on rate encoding and temporal encoding, which are essential for interpreting neural signals and implementing them in computational models. Neurons transmit information about the intensity or strength of a stimulus by varying the frequency of their spike trains. Higher firing rates correspond to stronger stimuli or more intense signals.

3.1 Rate encoding

Rate encoding, also known as frequency encoding, posits that the information is represented by the firing rate of the neuron; that is, the number of spikes per unit time. This encoding scheme is inspired by observations in biological neural systems where the intensity of a stimulus can be encoded by the rate at which a neuron fires. Neurons transmit information about the intensity or strength of a stimulus by varying the frequency of their spike trains. Higher firing rates correspond to stronger stimuli or more intense signals [17]. Rate encoding is widely used in models that require a direct relationship between stimulus intensity and neural response. However, it is less effective in capturing the precise timing of events, which can be critical in processes like sensory information processing and motor control.

Count rate encoding is the simplest form of rate encoding, where information is encoded in the total number of spikes that occur within a fixed time window [18]. The intensity of the input signal is directly proportional to the number of spikes; a stronger input generates more spikes, and a weaker input generates fewer spikes. This method is easy to implement and understand, making it a popular choice for initial experiments with SNNs.

Density rate encoding refines the basic concept of count rate encoding by focusing on the density of spikes over time, without being bound to a fixed counting window [13]. In this method, the firing rate is determined by the proximity of the spikes to each other, with a higher density of spikes representing a stronger signal intensity.

Population rate encoding extends the concept of rate encoding in multiple neurons or a population of neurons [19]. Instead of relying on the firing rate of a single neuron, population rate encoding considers the collective

firing rate of a group of neurons to represent the input signal. This method can encode a wider range of information using the diversity of response patterns within the population of neurons. Each neuron in the population may be sensitive to different aspects or features of the input signal, allowing the network to capture more complex patterns and nuances. Population rate encoding is particularly useful in networks that process multidimensional data or require a high degree of information specificity.

3.2 Temporal encoding

Temporal encoding strategies rely on the timing between spikes (inter-spike intervals) to convey information. This encoding method is more nuanced and can represent information in the precise timing of spikes, not just their rate. Temporal encoding suggests that the information content of a neural signal is contained in the specific timing of spikes. This can involve patterns of spikes that occur in rapid succession or the timing between individual spikes in relation to external events [20].

The most basic temporal coding scheme is the time-to-first-spike (TTFS), which encodes the magnitude or other relevant properties of a signal by emphasizing the timing of the first spike after the onset of a stimulus. This method is inspired by observations in biological neural systems, where the timing of neuronal spikes often conveys important information about sensory input or neural processing.

In TTFS encoding, the intensity or value of an analog signal is inversely proportional to the time at which the first spike is generated. Given the normalized signal amplitude X_i with each real value normalized, the time difference between the first spike and the onset of the stimulus Δt is shown as $\Delta t = 1/X_i$ or $\Delta t = 1 - X_i$ which means that the stronger or more intense the signal, the shorter the latency to the first spike, while the weaker the signal, the longer the latency.

The latency encoding method represents the intensity of an input signal through the timing of spike occurrences: the more intense the signal, the quicker the spikes are generated. This property allows latency encoding to accurately represent the temporal dynamics of a signal, making it highly suitable for tasks that require timely precision [21].

The simplest latency encoding is the linear latency encoding, which establishes a direct proportionality between the spike's timing and the signal's intensity. In practice, this means a signal that is twice as potent will trigger a spike in half the anticipated time. While this model is straightforward and easy to comprehend, it lacks differential sensitivity across the spectrum of signal intensities.

Power latency encoding relates the signal intensity to the timing of spikes by employing a power law [22]. This modification permits the expedited conveyance of more critical information compared to the linear approach.

The interspike interval (ISI) encoding is a method used to represent information based on the time intervals between consecutive spikes [23]. This encoding technique is particularly relevant in the context of time-sensitive neural processing, where the timing of spikes carries significant information. ISI encoding is a way to map the amplitude or features of an analog signal into the temporal domain, translating the signal's characteristics into a sequence of spikes with varying intervals.

In the TTFS approach, the occurrence of an input sets the timing for the first spike, with subsequent spikes encoded on the basis of their temporal deviation from this initial spike. Conversely, latency encoding assigns spike timings based on the intensity of the input signal. Meanwhile, the ISI encoding is characterized by the temporal intervals between each spike in a spike train.

The Send-on-Deluxe (SOD) spike encoding method is part of a broader category of methods known as event-driven or data-driven encoding strategies, which aim to efficiently represent information in a way that mimics the operation of biological neural systems. The core idea behind SOD is to generate spikes only when the change in the value of the input signal exceeds a certain threshold (ΔSOD) [24]. In SOD encoding, spikes are not generated at regular intervals or based on the absolute value of the signal. Instead, spikes are emitted based on the change in the signal's value, which can significantly reduce the number of spikes needed to represent a signal, thereby making the representation more efficient.

Recent studies have explored models that integrate rate and temporal encoding, aiming to harness the strengths of both approaches. These hybrid models can adapt dynamically to different types of stimuli and tasks, offering flexibility and robustness in information processing [25]

4 Learning rules of SNNs

Training SNNs is a multifaceted challenge that bridges the gap between biological inspiration and computational efficiency. This expanded section explores in depth the methodologies of “Bio-plausible Learning” and “Gradient-based Learning,” illustrating their significance, applications, and recent advancements within each domain.

4.1 Bio-plausible learning

Bio-plausible learning draws directly from the mechanisms observed within biological neural systems, aiming to replicate these processes within SNNs to achieve a more natural and efficient form of computation.

Spike-timing-dependent plasticity (STDP) is a fundamental mechanism in the realm of SNNs, which is pivotal to

understanding how synaptic connections evolve over time in response to neuronal activity patterns. STDP modulates synaptic strength based on the timing of pre-and post-synaptic spikes, embodying the principle that "neurons that fire together, wire together" [26].

STDP is a biological learning rule observed in the brain that describes how the connections, or synapses, between neurons change in strength in response to the timing of their spikes. It is a form of synaptic plasticity, which is a fundamental mechanism for learning and memory in the brain. STDP is particularly interesting because it provides a basis for understanding how complex neural circuits learn to process and store information based on temporal patterns of activity.

The core principle of STDP is that the strength of a synapse is adjusted on the basis of the precise timing of the spikes of the pre-synaptic and post-synaptic neurons. There are two critical aspects to this rule:

Long-Term Potentiation (LTP): If a pre-synaptic neuron fires and causes a post-synaptic neuron to fire shortly afterward (i.e., the pre-synaptic spike precedes the post-synaptic spike within a narrow time window), the synapse between these neurons is strengthened. This process is known as LTP. It means that the pre-synaptic neuron is effectively contributing to the firing of the post-synaptic neuron, reinforcing this connection.

Long-Term Depression (LTD): Conversely, if the post-synaptic neuron fires before the pre-synaptic neuron (i.e., the post-synaptic spike precedes the pre-synaptic spike within a certain time window), the synapse is weakened. This process is known as LTD. It suggests that the pre-synaptic neuron's firing is not effectively contributing to the post-synaptic neuron's firing, leading to a weakening of this connection.

Here's an STDP-based Unsupervised learning method:

$$\begin{cases} \Delta w_{ij} = a^+ \times w_{ij}, & \text{if } t_j - t_i \leq 0 \\ \Delta w_{ij} = a^- \times w_{ij}, & \text{if } t_j - t_i \geq 0 \end{cases} \quad (1)$$

where i and j refer to post- and pre-synaptic neurons, respectively, t_i and t_j denote the corresponding spike times, Δw_{ij} is the modification of the synaptic weight, and a^+ and a^- are adjustable hyperparameters specifying the amount of change. This method provides a mechanism for adjusting the connections of neural networks according to the relative timing of spikes [27].

STDP has been extensively modeled and applied in various computational frameworks to enhance the performance of SNNs in tasks such as pattern recognition, decision-making, and sensory processing [28, 29]. For example, it allows SNNs to autonomously learn and extract features from input data, facilitating unsupervised learning applications [27]. Furthermore, STDP contributes to the formation of memory and learning processes by reinforcing causal relationships

between neuronal firing events, which is essential for developing adaptive neural networks [30].

Recent advances have expanded STDP's applications, integrating it with deep learning techniques to create hybrid models that leverage both temporal dynamics and hierarchical data representations [31]. These models have shown promising results in processing complex datasets, such as high-dimensional EEG signals and intricate speech patterns, demonstrating STDP's versatility and potential in advancing SNN research [32].

Despite its benefits, challenges remain in optimizing STDP for large-scale networks and real-world applications. Issues such as computational efficiency, scalability, and the integration of STDP with other learning rules are areas of ongoing research [33]. Addressing these challenges will be crucial for fully harnessing STDP's capabilities in SNNs and furthering their application in neuromorphic computing and artificial intelligence.

STDP stands as a cornerstone of learning in SNNs, embodying principles of bio-plausibility and temporal sensitivity. As the field progresses, further exploration and refinement of STDP will undoubtedly unlock new potentials for SNNs, paving the way for sophisticated neural models capable of tackling complex cognitive tasks.

4.2 Gradient-based learning

Gradient-based learning techniques have been innovatively adapted for SNNs, allowing these networks to benefit from the vast advancements in optimization algorithms developed for artificial neural networks.

Backpropagation Through Time (BPTT) is an extension of the traditional backpropagation algorithm, adapted for training recurrent neural networks (RNNs). It addresses the challenge of learning temporal dependencies in sequential data by unfolding the RNN over time, allowing gradients to be computed for each timestep. This enables the network to update its weights based on the temporal sequence of inputs and outputs, effectively learning from sequences where the context is crucial. BPTT traverses the network from the final output back to the initial input, calculating the gradient of the loss function at each step with respect to the weights, considering the entire sequence [34, 35].

Applying BPTT to SNN for effective learning involves overcoming the challenges posed by the inherently discrete and non-differentiable nature of spiking events [36]. This necessitates the development of novel approaches, such as surrogate gradient methods, which enable the gradient descent optimization by approximating the gradients of spike functions. These methods have been pivotal in advancing the training capabilities of SNNs, making it feasible to capture temporal dynamics and dependencies in data more efficiently than traditional neural networks. By adding BPTT

and surrogate gradients, the accuracy and efficiency of SNNs have gotten a lot better in a lot of different areas, from neuromorphic computing to processing sensory data in real time. However, this approach also introduces complexity in terms of computational demand and the need for precise simulation of neural dynamics over time, raising ongoing research questions about scalability, optimization, and the balance between model complexity and computational efficiency.

Recent advances have seen the development of error backpropagation algorithms tailored for SNNs. This methodology, crucial for refining synaptic weights based on error gradients, improves the performance of SNN in tasks such as classification and regression [31].

In the future, this field will probably focus on making BPTT in SNNs more computationally efficient, creating more reliable surrogate gradient methods, and exploring new architectures that can make better use of SNNs' temporal processing abilities. As this area of research evolves, it holds the promise of enabling more sophisticated, energy-efficient, and biologically plausible models of computation that can outperform traditional approaches in processing temporal and sensory data.

The introduction of surrogate gradient methods addresses the challenge of non-differentiable spike functions, enabling the application of gradient descent in SNNs [37]. This approach has significantly propelled the training of deep SNNs, making deep learning techniques accessible for spike-based networks.

Surrogate gradient methods are a great new way to train SNNs that solves the main problem of how to teach them when spiking events are not differentiable. Gradient-based learning algorithms, such as backpropagation, can be used with these methods because they create a smooth and differentiable approximation of the spike function. This strategic approach has unlocked the potential of SNNs in various complex computational tasks, especially those involving the processing of temporal information. The development and refinement of surrogate gradient techniques is ongoing, with research aimed at improving the accuracy, efficiency, and biological plausibility of these networks. Future directions include optimizing these surrogate functions for better performance, reducing computational overhead, and integrating them into more scalable and robust SNN architectures. This advancement holds promise for significant contributions to fields such as neuromorphic computing, real-time sensory data processing, and autonomous systems, where SNNs can offer superior efficiency and performance due to their bio-inspired design [38–40].

4.3 ANN to SNN conversion

The transformation of ANNs to SNNs is a fundamental stride in leveraging the temporally dynamic nature of SNNs

for efficient neuromorphic computing. This conversion is motivated by the need to implement neural models that are not only powerful in terms of processing capabilities but also energy efficient and capable of real-time interaction with the environment [41, 42].

ANN-to-SNN conversion involves reinterpreting the continuous activation functions of ANNs into discrete spikes over time, a process that fundamentally shifts the computational paradigm from a continuous to a spike-based framework. To achieve this, each neuron in the ANN is mapped to a spiking neuron in the SNN, where the firing rate of the spiking neuron corresponds to the activation level of the traditional neuron. This rate-based coding ensures that the temporal dynamics of spike trains can approximate the analog outputs of the ANNs under certain constraints [43].

Various strategies have been developed to convert ANNs to SNNs effectively. One prevalent method involves adjusting the weights and biases of the ANN so that its behavior can be emulated by an SNN without loss of functionality or performance. The synaptic weights are typically scaled based on the maximum activation levels observed in the pre-trained ANN, which helps in maintaining the dynamic range in the spiking domain [41].

Another technique focuses on threshold balancing where the firing thresholds of the neurons are adjusted to optimize the spike generation process, ensuring that the SNN's response mirrors that of the original ANN as closely as possible. This method often involves a normalization process where the thresholds or the weights of the spiking neurons are scaled, taking into account the statistics of neuron activations during the ANN's training phase [44].

The direct conversion of ANNs to SNNs presents unique challenges, primarily due to the non-differentiable nature of spike functions, which complicates the use of conventional backpropagation algorithms. To address this, researchers have employed surrogate gradients or developed differentiable approximations of the spike function that facilitate the gradient-based optimization of SNNs [37].

Moreover, the inherently sparse and binary nature of spikes leads to information loss when converting from a high-resolution analog signal (in ANNs) to a binary format (in SNNs). Advanced strategies such as soft reset mechanisms, where the neuron's membrane potential is only partially reset after a spike, have been proposed to mitigate this loss, enhancing the fidelity of the conversion process [44].

5 SNN for speech signals

The utilization of SNNs for processing and recognizing speech signals represents a significant advancement towards creating more biologically plausible models of human auditory processing. SNNs, often described as the third

generation of neural network models, mimic the way biological neurons process information, making them uniquely suited for handling temporal data such as speech signals. This section reviews recent developments in the application of SNNs to speech signal processing, drawing on insights from key research papers.

Wu et al. introduced a biologically plausible speech recognition framework that leverages the dynamic and temporal processing capabilities of SNNs. Their work demonstrates how SNNs can effectively process speech signals by mimicking the auditory processing in the human brain, achieving notable results in speech recognition tasks [45].

Expanding on this, Ramírez-Mendoza et al. proposed a new spike membership function for the recognition and processing of spatiotemporal spike patterns, with a focus on syllable-based speech recognition. This innovative approach underlines the flexibility of SNNs in capturing the intricate patterns within speech signals, offering improvements in recognition accuracy and processing efficiency [46].

Hamian et al. introduced a novel learning approach that integrates deep SNNs with multi-objective optimization algorithms for automatic digit speech recognition. Their findings underscore the potential of SNNs in enhancing the performance of speech recognition systems through the application of advanced optimization techniques [47].

In the realm of feature extraction from speech signals, Tavanaei and Maida developed an SNN that learns to extract spike signatures from speech, showcasing the network's ability to identify distinctive features within speech signals that are crucial for accurate recognition [48].

Bittar and Garner contributed to the field by establishing a surrogate gradient spiking baseline for speech command recognition. Their research highlights the adaptability of SNNs to speech command tasks and the potential for these networks to operate efficiently in real-time applications [49].

Furthermore, advancements in SNN architectures for speech processing have been demonstrated by Sun et al. who introduced adaptive axonal delays in feedforward SNNs to enhance spoken word recognition accuracy. This approach illustrates the ongoing efforts to refine the temporal dynamics of SNNs for improved speech signal processing [50].

Pan et al. focused on auditory neural encoding and decoding within SNNs, presenting an efficient and perceptually motivated algorithm. Their work provides insight into how SNNs can be optimized to mimic the processing capabilities of the human auditory system, offering a path toward more naturalistic speech recognition systems [51].

In summary, the application of SNNs to speech signal processing has seen considerable progress, with researchers exploring various aspects of SNN architecture, learning algorithms, and application to specific speech recognition tasks. These advances highlight the potential of SNNs not only to mimic the biological processes underlying human

auditory perception, but also to improve the performance and efficiency of speech recognition systems. The studies reviewed here contribute to a growing body of evidence that SNNs offer a promising avenue for the development of advanced biologically inspired speech processing models.

Even though SNNs have come a long way for speech signals, there have been few benchmark audio datasets in pre-coded spiking form [43]. Some examples of neuromorphic audio benchmark datasets are shown in Table 1. These datasets include the N-TIDIGITS dataset [52], which recorded TIDIGITS through a Dynamic Audio Sensor (DAS), the SHD dataset, which recorded Heidelberg Digits as spikes through a cochlear model, and the SSC dataset, which recorded speech commands with the same model [53].

6 SNN for neurophysiological signals

The exploration of SNNs for the analysis of physiological signals represents a burgeoning area of research, harnessing the temporal dynamics and energy efficiency of SNNs to process and classify complex biological signals such as electromyograms (EMG), electrocardiograms (ECG), and electroencephalograms (EEG). This section delves into recent advancements in the application of SNNs for physiological signal processing, highlighting key contributions and findings from the literature.

Regarding research on EMG using SNNs, Tieck et al. showcased the application of SNNs in triggering robot hand reflexes using human EMG data. This study not only highlights the biocompatibility of SNNs but also their potential in prosthetics and rehabilitation devices [54].

Du et al. focused on continuous motion decoding of the upper limb using EMG signals and SNNs. Their work contributes to the development of more intuitive and responsive control systems for prosthetic limbs, leveraging the temporal sensitivity of SNNs [55].

In a study by Sun et al., the feasibility of applying SNNs in myoelectric control systems was examined, suggesting that SNNs can significantly improve the performance of myoelectric prostheses and other assistive technologies [56].

Table 1 Neuromorphic audio datasets

Audio datasets	Total speakers	Total dataset	
N-TIDIGITS [52]	225	4950	Dynamic Audio Sensor
SHD [53]	12	10,420	Cochlea Model
SSC [53]	1864	105,829	Cochlea Model
Spike-TIMIT [51]	630	6300	Biologically plausible Auditory Encoding

Donati et al. discriminated EMG signals using a neuromorphic implementation of an SNN, paving the way for the development of more compact and energy-efficient wearable devices for muscle activity monitoring [57].

In the realm of ECG anomaly detection, Gerber et al. implemented a neuromorphic system using delay chains to identify ECG irregularities. Their work underscores the capability of SNNs to perform real-time anomaly detection, offering a promising approach for continuous health monitoring systems [58].

Amirshahi and Hashemi developed an ECG classification algorithm using STDP and Reward-STDP (R-STDP) neural networks for real-time monitoring on ultra-low-power wearable devices. Their work demonstrates the potential of SNNs to provide efficient and accurate real-time monitoring of cardiac health, a critical step towards the development of next-generation wearable health monitoring systems [59].

Huang et al. presented an on-device, edge-learning framework for cardiac abnormality detection using a bio-inspired and spiking shallow network. This approach emphasizes the utility of SNNs in edge computing applications for healthcare, enabling local processing and immediate response, which is vital for patient monitoring and emergency detection [60].

Chu et al. developed a neuromorphic processing system with spike-driven SNN processor for wearable ECG classification. Their system showcases the potential of SNNs in wearable technology, offering a path towards more autonomous, efficient, and responsive healthcare devices [61].

Corradi et al. achieved real-time ultra-low power ECG anomaly detection using an event-driven neuromorphic processor. This work highlights the efficiency of SNNs in processing cardiovascular signals, offering solutions for long-term, continuous health monitoring with minimal energy consumption [62].

Feng et al. built and trained a deep spiking neural network for ECG classification, marking a significant step towards the integration of deep learning architectures with the temporal dynamics of SNNs for enhanced accuracy in cardiac signal analysis [63].

Xing et al. focused on accurate ECG classification using a Spiking Neural Network and an attentional mechanism for real-time implementation on personal portable devices. Their work underscores the enhancement of SNNs through attention mechanisms, improving the precision and reliability of ECG analysis in wearable technology [64].

Yan et al. explored energy-efficient ECG classification with a spiking neural network, addressing the critical issue of power consumption in wearable devices. Their findings contribute to the development of ultra-low-power, highly efficient wearable devices for long-term heart monitoring, highlighting the energy efficiency of SNNs [65].

For pursuing the versatility of EEG signals, a variety of works based on SNNs have been presented, recently.

The research by Shan et al. on compact seizure detection using SNNs and support vector machines for neuromorphic implementation illustrates the potential of SNNs in medical diagnostics, particularly in detecting neurological abnormalities from EEG signals with high accuracy and low power consumption [66].

Tian et al. introduced NeuroCARE, a neuromorphic edge computing framework for healthcare applications, showcasing the broad applicability of SNNs in healthcare, from real-time monitoring to predictive analytics [67].

Li et al. introduced a Fractal Spiking Neural Network scheme for EEG-based emotion recognition, presenting a novel approach to understanding and interpreting human emotions through physiological signals. Their work exemplifies the potential of SNNs in neuropsychological research and the development of human-computer interaction technologies, facilitating a deeper understanding of emotional states through EEG analysis [68].

Chen et al. conducted a feasibility study on the application of SNNs in myoelectric control systems, demonstrating the potential of SNNs to improve the performance of myoelectric prostheses and assistive technologies. Their work indicates the adaptability of SNNs to various signal types, including EEG, for enhancing the control and responsiveness of prosthetic devices [69].

Xu et al. introduced EESCN, a novel SNN method for EEG-based emotion recognition. Their approach highlights the capability of SNNs to process and classify EEG signals related to human emotions, offering insights into the development of more intuitive and responsive human-computer interaction systems [70].

Kumar et al. showcased the decoding of EEG signals using SNNs on neuromorphic hardware, emphasizing the efficiency and potential of SNNs in processing EEG data for BCI applications. Their research underscores the suitability of SNNs for real-time, low-power EEG signal processing, contributing to the advancement of neurotechnology and BCIs [71].

Honzík and Mouček explored the classification of BCI and image data using SNNs, demonstrating the versatility of SNNs in interpreting EEG signals for various applications, including BCI and neuroimaging. Their findings illustrate the potential of SNNs to enhance the accuracy and efficiency of data classification in neuroscience research and applications [72].

Kumarasinghe et al. delved into brain-inspired SNNs for decoding muscle activity and kinematics from EEG signals during hand movements. Their study provides a foundation for understanding the neural mechanisms of movement and the potential for SNNs to interpret EEG signals for prosthetic control and rehabilitation therapies [73].

Cai et al. proposed a bio-inspired spiking attentional neural network for attentional selection in the listening brain, leveraging EEG signals. This work contributes to the understanding of auditory attention and the development of BCIs that can adapt to the user's focus, enhancing the natural interaction between humans and machines [74].

Liao et al. presented a convolutional spiking neural network with adaptive coding for motor imagery classification, highlighting the application of SNNs in interpreting EEG signals for motor imagery tasks. Their research points to the potential of SNNs in developing more accurate and efficient BCIs for rehabilitation and virtual interaction [75].

Zhang et al. explored the use of non-iterative SNNs with attention mechanisms for motor imagery EEG classification. Their methodology demonstrates the effectiveness of incorporating attention mechanisms within SNNs, enhancing the accuracy of brain-computer interface applications [76].

Collectively, these studies underscore the versatility and efficiency of SNNs in processing physiological signals. From wearable technology for health monitoring to advanced prosthetics and brain-computer interfaces, SNNs offer a biologically inspired approach that mimics the brain's processing capabilities, opening new avenues for real-time, energy-efficient, and highly accurate physiological signal analysis.

7 Conclusions and future perspectives

The comprehensive exploration of SNNs in processing physiological and speech signals marks a significant advancement in the realms of neurotechnology and artificial intelligence. This review has highlighted the pivotal role that SNNs play, leveraging their unique capabilities to mimic the processing mechanisms of the human brain, thereby offering a novel approach to understanding and interpreting complex temporal data. Through detailed analysis, we have observed how SNNs provide substantial benefits in applications ranging from real-time health monitoring systems to sophisticated brain-computer interfaces and prosthetic controls, promising a new era of personalized medicine and enhanced human-computer interaction.

Notably, several transformer-based SNNs have been developed, providing new insights for advanced SNNs. Spikformer [77] designed an innovative SNN utilizing spiking self-attention (SSA) by using Query, Key, and Value in spike-form without the softmax function. More recent work, i.e., SGLFormer [78] integrated convolutional structure and transformer structure in SNN form, resulting in a directly trained SNN. However, those transformer-based SNNs are still limited in using visual datasets, thus extensive application to speech and physiological signals needs to be investigated.

Challenges remain, particularly in the development of efficient learning algorithms, the scalability of hardware to support SNN models, and the integration of these networks into practical real-world applications. Addressing these challenges will require a concerted effort from the research community to innovate and refine SNN technologies.

Future research directions are poised to further harness the potential of SNNs, focusing on the creation of more advanced SNN architectures, the development of robust and scalable training methodologies, and the exploration of hybrid computational models that combine the strengths of SNNs with other neural network frameworks. Such advances could significantly impact various sectors, including healthcare, where they could lead to breakthroughs in patient monitoring and treatment, and in technology, by creating more naturalistic interfaces for human-computer interaction.

As we continue to bridge the gap between biological neural systems and artificial intelligence, SNNs stand at the forefront of this exciting frontier, promising to unlock new capabilities and applications that were previously unattainable. The journey ahead is filled with opportunities to revolutionize our approach to computing, medicine, and beyond, making this an incredibly promising time for researchers and practitioners in the field.

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Declarations

Conflict of interest The authors have no relevant financial or nonfinancial interests to disclose.

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